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Wearable Healthcare devices include smart watches, fitbit wristbands, fitness trackers, helmet-worn impact checkers, back pain posture checkers, necklaces, clip-ons, smart clothing and filament implant. These medical devices with select sensors can record real-time data on various physiological conditions and electronically send them to doctor, or via apps on smart phone. It has two main fitness apps such as **Activity app and Work out app.** Activity app is all about health, movement, wellness and daily routine; and **Workout app** tracks running, cycling and walking. The wearable healthcare devices are activity trackers, which offer health benefits by measuring number of steps, distance travelled, calories burned, heart rate, detects blood volume and capillary-size changes under pressure, body temperature and quality of sleep. It also tracks progress over time so that one can set daily goals for bodyweight, blood glucose level, blood pressure, sleep time, water intake, etc.

necessary controls such as temperature of reaction, volumes of sample, reagents and transfer of fluids. More sophisticated multichannel autoanalysers with capabilities of selecting any type of analysis for any one set of samples are available now (Figure12.4).

12.2. Imaging Instruments

EEG – Electroencephalogram

EEG is a test to evaluate the electrical activities of the brain (Figure12.5). Brain cells communicate with each other through electrical impulses. An EEG tracks and records the brain wave patterns. It is a graphical recording of the electrical activities of the cortical and sub cortical regions of the brain. It is recorded by placing the surface electrodes on the scalp region. The electrodes analyse the electrical impulses in the brain and send signals to a computer which records the results. In 1929 Germen scientist Hans Berger was the first to analyse the EEG. Hence, EEG is also known as "Berger Wave". The electrical

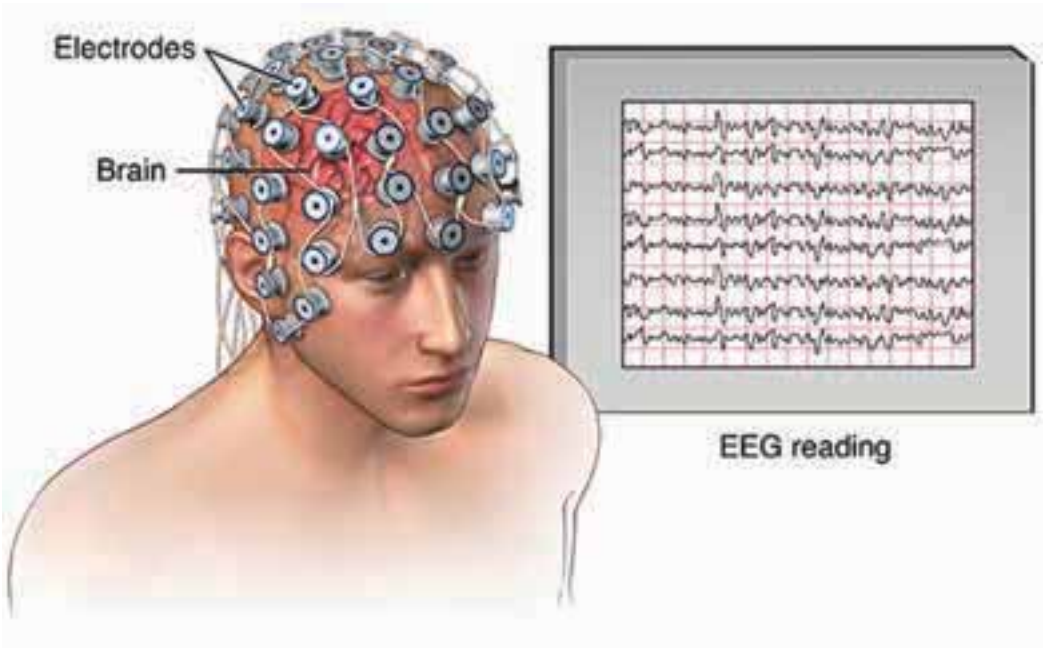

Figure 12.5 EEG with waves